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A Comparative Study of Node Ranking Metrics in Directed Networks

Master Project

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Abstract

This project presents a comparative study of three node ranking metrics – in-degree, PageRank, and Zermelo’s strength – in competitive directed networks. Through numerical simulations on different synthetic networks, the study evaluated their mutual correlations with respect to the predefined intrinsic strength of the nodes. Correlation analyses using Pearson and Spearman coefficients reveal that Zermelo’s strength provides the most accurate estimates across all network configurations, particularly in networks where participation is unequal. In-degree shows high sensitivity to network density and participation bias, while PageRank exhibits moderate performance with limited dependence on its teleportation probability parameter. This work is a basis for future analysis of real-world competitive data in professional tennis rankings.

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1 Introduction

The study of rankings in competitive networks is a standard problem in network science, with applications ranging from sports tournaments to social hierarchies. In the context of directed graphs, edges represent the outcomes of pairwise interactions, such as victories in tournaments or citations in scientific publications. Assigning a numerical value to each participant based on these results allows for comparison across the system.

1.1 Historical Background

Several methods have been proposed to rank nodes in directed networks. Among the earliest is Zermelo’s method, introduced in 1929 [1], which models tournament outcomes as a probability maximisation problem. Inspired by similar reasoning, the probabilistic Bradley-Terry (BT) model followed in 1952 [2]. It assigns intrinsic strengths to each competitor, from which the probability of winning can be derived. The corresponding Zermelo’s algorithm provides an iterative method to estimate them from observed results. Recently, researchers have developed more efficient algorithms that converge faster to the same stable solution [3]. Moreover, extensions of the model have included realistic factors, such as a luck component and depth of competition, leading to improved predictions [4].

Another widely used approach is PageRank (PR), developed by Larry Page and Sergey Brin in 1999 [5]. It was initially designed to rank web pages by evaluating their importance according to the pages linking to them. Beyond its application to web search, it has been applied to social networks and competitive systems, including sports. However, recent studies indicate that PageRank may be limited in certain contexts, such as competitive sports, where it does not always provide the most accurate rankings [6].

As a baseline, in-degree is a simple metric that counts the number of incoming edges for each node. It provides a point of comparison for more complicated ranking methods.

1.2 Objectives of the Work

The main goal of this work is to analyse different ranking methods in competitive networks. The study focuses on understanding how approaches such as Zermelo’s strength model and PageRank assign scores to nodes and differ in their rankings. Through numerical simulations on synthetic data across different types of networks, correlations between the different metrics and the actual node strengths are computed, allowing evaluation of the accuracy and limitations of each method.

This project serves as a first step towards a master thesis that will extend the analysis to real-world data from professional tennis. The objective will be to design a mathematical model that reproduces patterns similar to real ATP results and evaluate the performance of ranking methods in this specific context.

2 Theoretical background

2.1 Model of the Directed Network

Competitive systems are modelled using directed graphs composed of N nodes and E edges. Each node represents a participant in the competition, such as a player in a tennis tournament. A directed edge from node i to node j represents a victory of participant j over participant i . Formally, a binary variable e_{ij} is defined as

$$e_{ij} = \begin{cases} 1 & \text{if there exists a directed edge from } i \text{ to } j, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Each pair of distinct nodes can be connected by at most one edge in a given direction. Hence, the maximum number of edges in the network is $E_{max} = N(N - 1)$. The density of the network, denoted by d , is defined as the ratio of the number of existing edges E to E_{max} :

$$d(E, N) = \frac{E}{N(N - 1)}. \quad (2)$$

This quantity provides an indication of how densely the network is connected.

This model is a simple directed graph, which allows at most one edge between any pair of nodes in a given direction. In reality, participants can compete multiple times in competitions, such as tennis or football leagues. This can be modeled with a multigraph, where the variable e_{ij} is replaced by w_{ij} , representing how many times participant j defeated participant i . This approach will not be used in this project, but will be relevant for the analysis of tennis data in the following Master thesis.

2.2 Node Metrics

Different approaches can be used to evaluate the performance of nodes in a directed network. Among the existing metrics, three are considered in this project: the in-degree, the PageRank score and Zermelo's strength.

2.2.1 In-Degree

The in-degree of a node is defined as the number of edges directed towards that node. In the context of competitions, it represents the number of wins by one participant over others. Mathematically, the in-degree s_i^{in} of node i is given by

$$s_i^{in} = \sum_{j \neq i} e_{ji}. \quad (3)$$

This metric gives an intuitive way of having a measure of the performance, although it does not take into account the strengths of the opponents.

On the other hand, the out-degree of a node s_i^{out} is defined as the number of edges leaving that node:

$$s_i^{out} = \sum_{j \neq i} e_{ij}. \quad (4)$$

2.2.2 PageRank Score

The PageRank algorithm ranks nodes according to their importance in the directed network. It assigns a score to each node that accounts not only for the number of incoming edges, but also for the importance of the nodes from which those edges originate. In other words, a node linked by highly ranked nodes receives more influence. Formally, the PageRank P_i of node i is defined as

$$P_i = (1 - \alpha) \sum_{j: s_j^{out} > 0} P_j \frac{e_{ji}}{s_j^{out}} + \frac{\alpha}{N} + \frac{1 - \alpha}{N} \sum_{j: s_j^{out} = 0} P_j, \quad (5)$$

where α is the teleportation probability ($0 < \alpha < 1$) [6]. It controls the balance between contributions from the network structure and random jumps. It represents the proportion of the total score that is redistributed uniformly to all nodes at each iteration, ensuring that every node receives some influence regardless of its connectivity.

The first term corresponds to contributions from incoming neighbours, weighted by their out-degree. Each node j distributes its score equally among all nodes it has lost to. Thus, a victory against a strong opponent with few defeats contributes more to the score of node i than a weaker opponent with many defeats. The second term represents a uniform redistribution of part of the total score across all nodes. The third term redistributes the score of nodes with no outgoing edges (called dangling nodes) evenly across all nodes in the network. It ensures that no score is lost during the calculation, even when some nodes have zero out-degree.

Note that the total score of all nodes is normalized to 1 for all values of α , ensuring that the sum of PageRank scores remains constant. For the iterative calculation, the initial values are set uniformly:

$$P_i^{(0)} = 1/N, \quad \forall i. \quad (6)$$

2.2.3 Zermelo's Strength

Zermelo's method is based on the Bradley-Terry model [2], which assumes that each node i has a positive strength parameter π_i . The probability that player i defeats player j is then given by

$$p_{ij} = \frac{\pi_i}{\pi_i + \pi_j}. \quad (7)$$

To estimate these strengths $\pi = (\pi_1, \dots, \pi_N)$ from the observed outcomes stored in the matrix $E = (e_{ij})_{1 \leq i, j \leq N}$, the maximum-likelihood estimate (MLE) can be used [3]. First, the likelihood of observing E given the strength vector π is

$$P(E | \pi) = \prod_{i \neq j} \left(\frac{\pi_i}{\pi_i + \pi_j} \right)^{e_{ij}}. \quad (8)$$

The corresponding log-likelihood can then be differentiated with respect to π_i for any i , and the derivative is set to 0:

$$\frac{\partial}{\partial \pi_i} \log P(E | \pi) = \frac{1}{\pi_i} \sum_j e_{ij} - \sum_j \frac{e_{ij} + e_{ji}}{\pi_i + \pi_j} = 0. \quad (9)$$

This equation can be rearranged into an explicit form, which defines Zermelo's algorithm [1]. At each iteration t , the updated strength $\pi_i^{(t+1)}$ is computed as

$$\pi_i^{(t+1)} = \frac{\sum_j e_{ji}}{\sum_j (e_{ij} + e_{ji}) / (\pi_i^{(t)} + \pi_j^{(t)})}. \quad (10)$$

By rearranging the terms in equation (9) differently, a faster version, proposed by Newman [3], can also be used:

$$\pi_i^{(t+1)} = \frac{\sum_j e_{ji} \pi_j^{(t)} / (\pi_i^{(t)} + \pi_j^{(t)})}{\sum_j e_{ij} / (\pi_i^{(t)} + \pi_j^{(t)})}. \quad (11)$$

In practice, the algorithms start with random initial strengths $\pi_i^{(0)} = e^{s_i}$, where the s_i are drawn from a logistic distribution:

$$f(s) = \frac{1}{(e^{-s} + 1)(e^s + 1)}. \quad (12)$$

This choice follows the methodology used by Newman [3]. Other initial distributions can be considered, such as a uniform one or an initialisation proportional to the in-degree of the nodes (with an offset to avoid zeros). All choices are expected to converge to the same solution, possibly with different convergence speeds.

The estimated strengths are updated asynchronously, meaning that the new value of each node is used immediately for the following updates. The solution is reached when the total variation across all nodes satisfies

$$\sum_i |\pi_i^{(t+1)} - \pi_i^{(t)}| < N \cdot tol, \quad (13)$$

where tol is a tolerance parameter, typically $tol = 10^{-6}$. This iterative procedure converges in both cases to the MLE $\hat{\pi}$ of the true strengths π . Because it maximises the likelihood of the observed outcomes, the estimates are consistent.

Since the BT model is invariant under multiplication of all strengths by a positive constant, the final estimations are normalized such that $\prod_i \pi_i = 1$. This way, the solution obtained is guaranteed to be unique.

2.3 Network Generation Models

To test and compare the accuracy of ranking metrics, numerical simulations are conducted on three distinct types of directed networks. Using synthetic data allows control over the structure and competitive outcomes, enabling the study of metrics under different conditions. For the computation of Zermelo’s strengths, it is important that the network is sufficiently connected, meaning that there is a directed path connecting most pairs of nodes. This ensures that the computed strengths are well-defined and avoids isolated nodes whose values could become zero. In the context of tennis, the main requirement is that each participant takes part in enough matches so that the ranking algorithm produces stable and meaningful estimates.

2.3.1 Random Network

Simple random networks are generated by choosing E directed edges uniformly at random from the set of all possible edges among N nodes. Self-loops are not allowed, as a player cannot compete against themselves in a match.

2.3.2 Bradley-Terry Network

A Bradley-Terry network is a directed network generated according to the BT model introduced in Section 2.2.3. Each node is assigned a true strength drawn from a uniform distribution between 0 and 1, representing its ability. These strengths establish an inherent hierarchy among the nodes, determining probabilistically which participants are likely to win or lose. Pairs of nodes are then selected at random, allowing the same pair to be drawn multiples times. For each selected pair (i, j) , the outcome of the match is decided using the BT probability p_{ij} defined in Equation (7). An edge is added from the loser to the winner to represent the result in the graph.

This process is repeated until E matches have been simulated. Since the network is represented as a simple directed graph, if the same pair meets again and the outcome is identical, no new edge is created. However, if the outcome is reversed in a subsequent match, an additional edge in the opposite direction is also created. Consequently, the final number of edges may be slightly lower than the total number of simulated matches.

2.3.3 Biased Network

A biased network is generated similarly to a BT network, with the difference that pairs of nodes are not selected uniformly at random. Instead, nodes with lower indices are more likely to be chosen, with probability for each node i to be selected given by

$$P(i) = \frac{1/\sqrt{i+1}}{\sum_{j=0}^{N-1} 1/\sqrt{j+1}}, \quad i = 0, \dots, N-1. \quad (14)$$

Once a pair of nodes is selected according to these probabilities, the outcome of the match is determined from the relation (7) with predefined strengths, and directed edge is added from the loser to the winner. The process is repeated until E matches have been simulated. As in the BT networks, repeated encounters between the same pair do not create multiple edges in the same direction, although edges in opposite directions may exist. This type of network introduces unequal participation, independently of their true abilities. It allows the study of ranking metrics under biased conditions, specifically due to some players having more games than others.

2.4 Correlation Analysis

To quantify the performance of the metrics under the different types of networks, a correlation analysis is conducted. For the simple random network, pairwise correlations between the three calculated metrics are studied to assess their mutual relationships. In contrast, the Bradley-Terry and biased networks are generated from a predefined set of true strengths. For these models, the accuracy of each metric is measured by directly comparing the computed scores with the true strengths.

To carry out this evaluation, two correlation coefficients are calculated: the Pearson coefficient, which quantifies the linear relationship between two variables, and the Spearman coefficient, which measures monotonic relations based only on their rank order. In other words, Pearson reflects how closely the numerical values of the data follow a linear trend, while the Spearman only focuses on the ordering of values.

Mathematically, the Pearson correlation coefficient between two variables X and Y is defined as

$$r_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (15)$$

where $\text{cov}(X, Y)$ is the covariance between X and Y , and σ_X, σ_Y their respective standard deviations [7].

On the other hand, the Spearman correlation coefficient is given by

$$\rho = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}, \quad (16)$$

with d_i representing the difference between the ranks of the corresponding variables x_i and y_i [8]. Alternatively, Spearman can be seen as a Pearson correlation applied to the ranks of X and Y rather than their actual values.

Both coefficients are defined between -1 and 1 . A value of 1 indicates a perfect positive correlation, meaning that increasing one variable also increases the other in a perfectly linear (Pearson) or monotonic (Spearman) way. Similarly, a value of -1 indicates a perfect negative correlation, while a 0 means no correlation.

3 Numerical Implementation

This section describes the work done to implement and compare the ranking algorithms presented in Section 2.2. All simulations were carried out in Python. The complete code is available in an online repository¹. It contains a Python notebook (.ipynb) with two data files (.pkl) storing results that can be directly reloaded to reduce computation time. Construction of directed graphs are based on the **NetworkX** library [9].

3.1 Generation of Synthetic Networks

To evaluate the performances of the ranking metrics, synthetic networks are generated as described in Section 2.3. Each type of network is represented as a directed graph, with the corresponding Python's `nx.DiGraph` object. Its structure is defined by two parameters: the number of nodes N and the number of edges E .

For the random network model, a function was first implemented from scratch to become familiar with the basics of the **NetworkX** library. For the main simulations, the predefined function `nx.gnm_random_graphs(N,E, directed=True)` was then used to generate such networks.

The BT networks were generated by assigning a true strength to each node and creating edges probabilistically according to the BT model (see Section 2.2.3). Similarly, the biased networks uses the same procedure, but nodes are chosen non-uniformly following the selection probabilities from equation (14).

3.2 Implementation of Ranking Metrics

The three ranking metrics introduced in Section 2.2 were implemented. To evaluate and compare their performances, tests were conducted on simple random networks with $N = 50$ nodes and $E = 1000$ edges. These specific values were chosen to ensure that the graphs are strongly connected and to avoid convergence issues in the algorithms. Results are stored in a dictionary for each metric, where each key corresponds to a node, and the associated value is the node's score.

3.2.1 In-Degree

The in-degree of nodes in a directed graph G was computed directly using the available function `G.in_degree()`. It simply iterates over all nodes and counts the number of incoming edges.

¹The full code and data files are available at: https://github.com/Theomarsy/M_project.git

3.2.2 PageRank Algorithm

Initially, a custom implementation of the PR algorithm was carried out iteratively based on equation (5), starting from an initial uniform distribution (see equation (6)). The convergence criterion used is similar to that defined for Zermelo’s strength in (17), i.e. the sum of the variations between two consecutive iterations t and $t + 1$ on all nodes must be less than N times a tolerance threshold tol :

$$\sum_i |P_i^{(t+1)} - P_i^{(t)}| < N \cdot tol, \quad (17)$$

To validate this implementation, the results were compared to the built-in function `nx.pagerank()` available [10]. It allows to verify whether the home-made implementation is correct. Note that the argument α_{NX} in `NetworkX` corresponds to one minus the teleportation probability α from equation (5) defined in the theory (i.e. $\alpha_{NX} = 1 - \alpha$). The comparison was made by calculating the mean absolute difference on all nodes between the two versions, with parameter $\alpha = 0.15$. Finally, the two versions were also compared in terms of execution time, by averaging the time required to reach convergence on 100 different graphs.

3.2.3 Zermelo’s Strength

Zermelo’s strength was implemented iteratively in two different ways: with Zermelo’s algorithm (10) and with the alternative version (11) proposed by Newman. The initial distribution of strengths and the convergence criterion is chosen as explained in the theory (see Section 2.2.3). A verification was then performed to ensure that both versions converge to the same solution after normalisation. The convergence speed was also studied to identify the most efficient version.

Moreover, the effect of the initial distribution on convergence speed was studied by comparing three options: the logistic distribution based on equation (12), the uniform distribution in $(0, 1]$ and an initialisation related to the in-degree s_i^{in} of each node i , given by

$$\pi_i^{(0)} = s_i^{in} + 1. \quad (18)$$

A +1 offset is added to ensure that no initial strength is zero, preventing division by zero during the algorithm. For each initial distribution, the mean number of iterations was computed over 100 random graphs, and the final estimated strengths were also compared. Note that for the rest of the project, the distribution described in theory is used.

3.3 Metrics Performance Evaluation

In this section, correlations are calculated either between metrics or with intrinsic strengths. To quantify them, Pearson and Spearman coefficients are used as defined in the equations

(15) and (16). They are directly computed from the available functions `pearsonr` and `spearmanr` from the `scipy.stats` library.

3.3.1 Random Networks

For random networks, the two correlation coefficients were calculated for each pair of metrics. In order to obtain relevant values, the results were averaged over 200 random graphs, each containing $N = 50$ nodes and $E = 1000$ directed edges. The metrics were computed for each graph. Then, the correlations were calculated to obtain an average with a standard error across all graphs. In the case of PageRank, the dependence on the teleportation probability α was also studied, considering values ranging from 0.05 to 0.95 in steps of 0.1. The values 0.99 and 0.999 were also introduced to check whether the correlations would behave unexpectedly. This precaution was taken because, for $\alpha = 1$, all PageRank scores are identical, making both coefficients undefined.

3.3.2 Bradley-Terry and Biased Networks

First, the correlation coefficients between the true strength of nodes and the metrics were calculated using the same parameters and procedure as for random networks. This first step enabled an initial analysis of the correlations for BT and biased networks.

In a second step, the analysis was extended to study the dependence of correlations on the size N of the network, as well as its density d defined in equation (2). The number of nodes N varies between 25 and 100 in steps of 25, while the density d was tested between 0.1 and 1 in increments of 0.1. It is important to note that the density requested in the arguments of the code does not correspond exactly to the actual density obtained. This is because the number of edges E required to get the correct density will not be reached, as a few directed edges may be repeated some times during generation of the graph. For visual representation of the results, the plots indicate this actual density rather than the requested one.

The main results were visualised using mainly three types of graphs. First, the correlations of the three metrics with true strength were plotted as a function of density for each N , considering only $\alpha = 0.15$ for PageRank. Next, each metric was plotted individually as a function of density to see the effect of N on the correlations. Finally, the correlations with PR were studied with different teleportation probability values to see if the performance of this metric depends heavily on the parameter α .

4 Results

This section focuses on presenting the main results obtained during the project, based on the methodology described in Section 3. Interpretation and discussion are left for the next section.

4.1 Validation of Algorithmic Implementations

Before studying the correlations between the different metrics and the true node strengths, it is important to check that the implemented algorithms are functioning correctly.

4.1.1 Comparison of Custom and Built-in PageRank Implementations

For PageRank, validation was performed by comparing the scores obtained for the same random network using the function implemented from scratch and `nx.pagerank()`. The results show that the mean absolute difference across all nodes between the two versions is zero. In terms of performance speed, the built-in function reaches convergence more quickly. The ratio between the average computation time for the home-made implementation and that of the integrated function is 5.16 ± 0.25 .

4.1.2 Comparison of Zermelo’s Strength Implementations

Zermelo’s strength was implemented according to Zermelo’s algorithm (slow) and to Newman’s new version (fast). The objective now is to verify that the two implementations converge to the same values, and which one converges faster to the solution. Figure 1(a) shows the values of the forces calculated with the fast algorithm on the x-axis and those calculated with the slow algorithm on the y-axis. The dotted line represents perfect equality between the strengths obtained ($x = y$). The average absolute difference in forces between the two algorithms is $(6.15 \pm 1.92) \cdot 10^{-7}$. Figure 1 (b) shows, for each algorithm, the total difference in strengths between two consecutive iterations as a function of the number of iterations. Note that the y-axis is on a logarithmic scale, and that the dotted vertical lines indicate the number of iterations required to achieve convergence for each implementation (i.e. the point at which the total difference is less than $N \cdot tol$).

The influence of the initial distribution of strengths on the final solution and on the convergence speed of Zermelo’s algorithm has also been studied. The three initial distributions were considered: logistic, uniform, and based on the incoming degree of the nodes. Table 1 shows the average number of iterations required to achieve convergence according to the algorithm and the initial distribution.

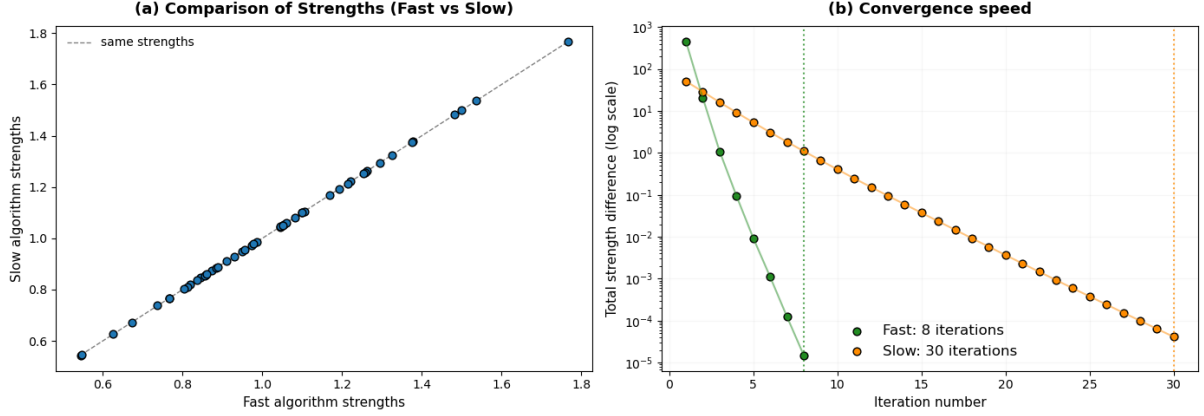


Figure 1: Comparisons of the two implementations for calculating Zermelo’s strengths (slow and fast versions) on a random directed network with $N = 50$ and $E = 1000$. (a) Node strengths with the fast algorithm versus the slow algorithm, with the dotted line corresponding to $x = y$. (b) Total difference between two successive iterations as a function of the number of iterations, with the vertical axis on a logarithmic scale.

Moreover, the average absolute differences between the strengths obtained according to the different initial distributions are grouped in Table 2. Figure 2 illustrates these results graphically. Subfigures (a) and (b) compare the strengths estimated with the fast algorithm from different initial distributions. Subfigures (c) and (d) show, for each algorithm, the total difference in forces between two consecutive iterations as a function of the iteration number, with the y-axis on a logarithmic scale. The vertical dotted lines indicate the number of iterations required to achieve convergence.

Table 1: Average number of iterations required to reach convergence, depending on the algorithm and initial distribution. Average iterations needed for the Zermelo’s computed strengths to converge, calculated over 100 simple random graphs ($N = 50$, $E = 1000$).

Algorithm	Initial distribution		
	Logistic	Uniform	In-degree
Fast	7.92 ± 0.39	6.98 ± 0.37	8.07 ± 0.41
Slow	30.13 ± 2.67	23.81 ± 2.09	30.96 ± 3.83

Table 2: Mean absolute differences between strengths obtained with different initial distributions. Values are averaged over 100 simple random graphs ($N = 50$, $E = 1000$).

Comparison	Mean absolute difference
In-degree vs Logistic	$(2.11 \pm 0.31) \cdot 10^{-8}$
In-degree vs Uniform	$(8.57 \pm 1.34) \cdot 10^{-8}$
Logistic vs Uniform	$(1.03 \pm 1.65) \cdot 10^{-7}$

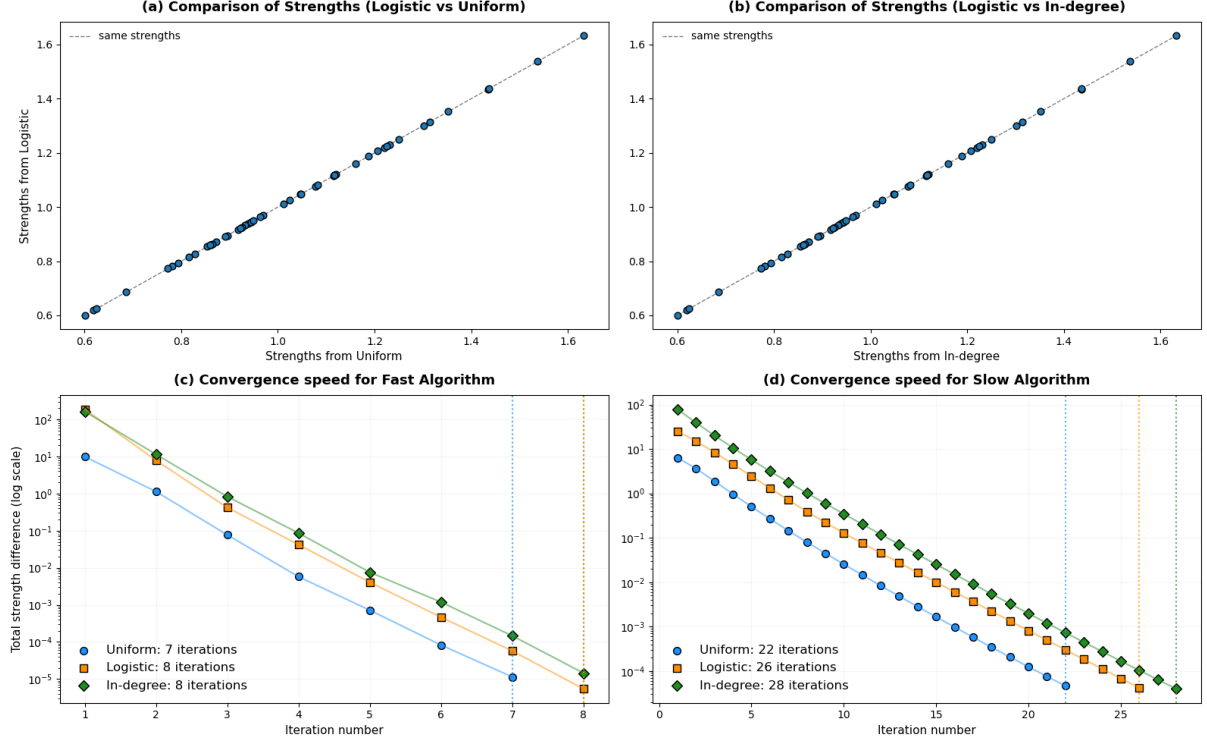


Figure 2: Comparison of the estimated Zermelo’s strengths calculated using different initial distributions and the convergence speed of the two implementation algorithms (fast and slow) on a random directed network with $N = 50$ and $E = 1000$. Comparison of estimated strengths (fast version) when initialised with a logistic versus a uniform distribution (a), and versus an in-degree-based distribution (b). Total strength difference between two successive iterations (on a logarithmic scale) for the fast (c) and the slow (d) algorithm, across the initial distributions.

4.2 Correlation Analysis of Metrics on Synthetic Networks

This subsection focuses on the results for the evaluation of Pearson and Spearman correlation coefficients between the different metrics, and between these metrics and the intrinsic strength of the nodes.

4.2.1 Inter-Metric Correlations

For random networks, correlations between pairwise metrics were computed for each teleportation probability value α when PageRank was considered. The correlation obtained between in-degree and Zermelo’s strength is 0.666 ± 0.006 according to Pearson and 0.679 ± 0.006 according to Spearman. The evolution of PageRank correlations with in-degree and Zermelo’s strength are shown in Figure 3(a) and Figure 3 (b), respectively, as a function of the teleportation probability α .

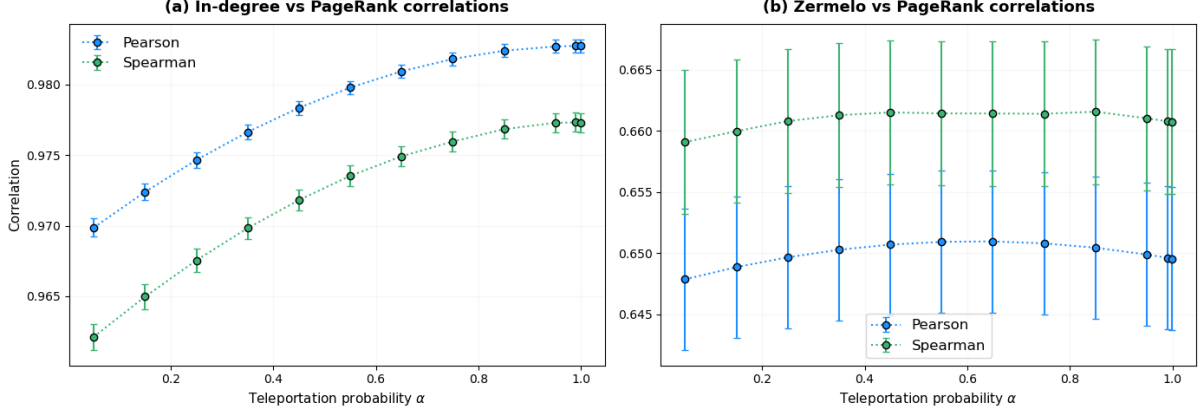


Figure 3: Correlations between PageRank and other metrics on random directed networks with $N = 50$ and $E = 1000$. (a) In-degree vs. PageRank. (b) Zermelo vs. PageRank. Both Pearson and Spearman correlations are shown. The values shown are averages over 200 graphs.

4.2.2 Correlations with True Strength

The correlations between metrics and true strengths were studied for Bradley-Terry networks and biased networks. Table 3 presents the Pearson and Spearman coefficients for the in-degree and the computed Zermelo’s strength. Figure 4 illustrates the evolution of correlations between PageRank and true strengths as a function of probability α with standard errors, separately for BT networks (a) and biased networks (b).

Table 3: Pearson and Spearman correlations between the true strengths and in-degree/Zermelo’s computed strengths for BT and biased networks. Correlations are averaged over 200 graphs, with $N = 50$ and $E = 1000$.

Correlation	BT networks		Biased networks	
	<i>Pearson</i>	<i>Spearman</i>	<i>Pearson</i>	<i>Spearman</i>
<i>In-degree vs True Strength</i>	0.779 ± 0.004	0.764 ± 0.005	0.514 ± 0.007	0.560 ± 0.007
<i>Zermelo vs True Strength</i>	0.796 ± 0.004	0.847 ± 0.003	0.731 ± 0.006	0.828 ± 0.003

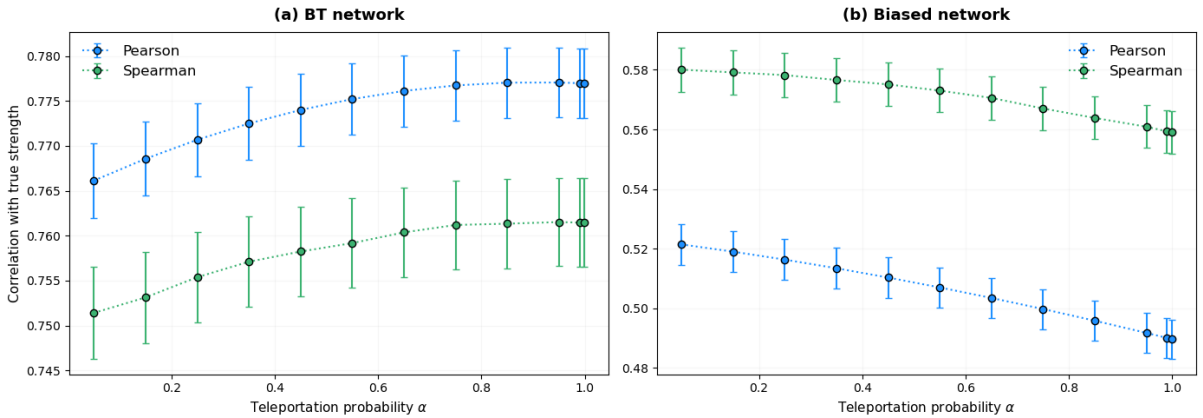


Figure 4: Correlations between PageRank and true intrinsic strengths on BT and biased directed networks with $N = 50$ and $E = 1000$. (a) BT networks. (b) Biased networks. Both Pearson and Spearman correlations are shown. The values shown are averages over 200 graphs.

4.3 Effect of Network Properties on Metric Correlations

Here, the effect of network properties on the correlation of metrics with the true strength of nodes is presented. Only Spearman's correlation is considered, as it has been observed in previous sections that Pearson's and Spearman's coefficients follow the same trends with a slight shift. Figure 5 shows the correlation of the three metrics with the intrinsic strength as a function of network density d for both types of networks. For each N , a graph was constructed to compare the performance of the three metrics as a function of network size and density. Figure 6 represents the same data, but displays the different curves for the different values of N simultaneously, with a graph for each metric and network type. It is therefore possible to see directly the influence of N on the correlation for the three metrics. For reasons of readability, the curve corresponding to $N = 75$ is not displayed. Finally, Figure 7 shows Spearman's correlation for PageRank, focusing on the influence of the parameter α . Only three teleportation probability values have been displayed (0.15, 0.55, 0.85). The results for other values have also been calculated, but only these are displayed in order to be able to compare the curves.

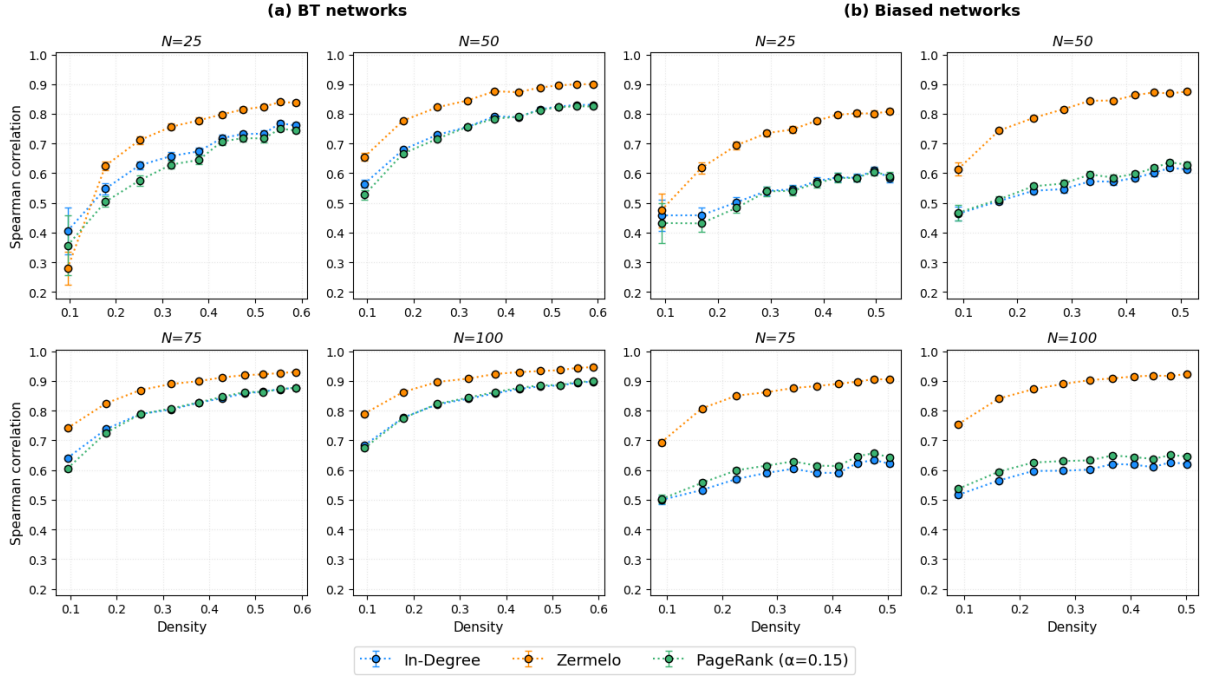


Figure 5: Spearman correlations between true intrinsic strengths as a function of density d . The subplots are grouped by network size N and compares the performance of in-degree, Zermelo and PageRank. Correlations are shown with their standard errors for $N = 25, 50, 75, 100$.

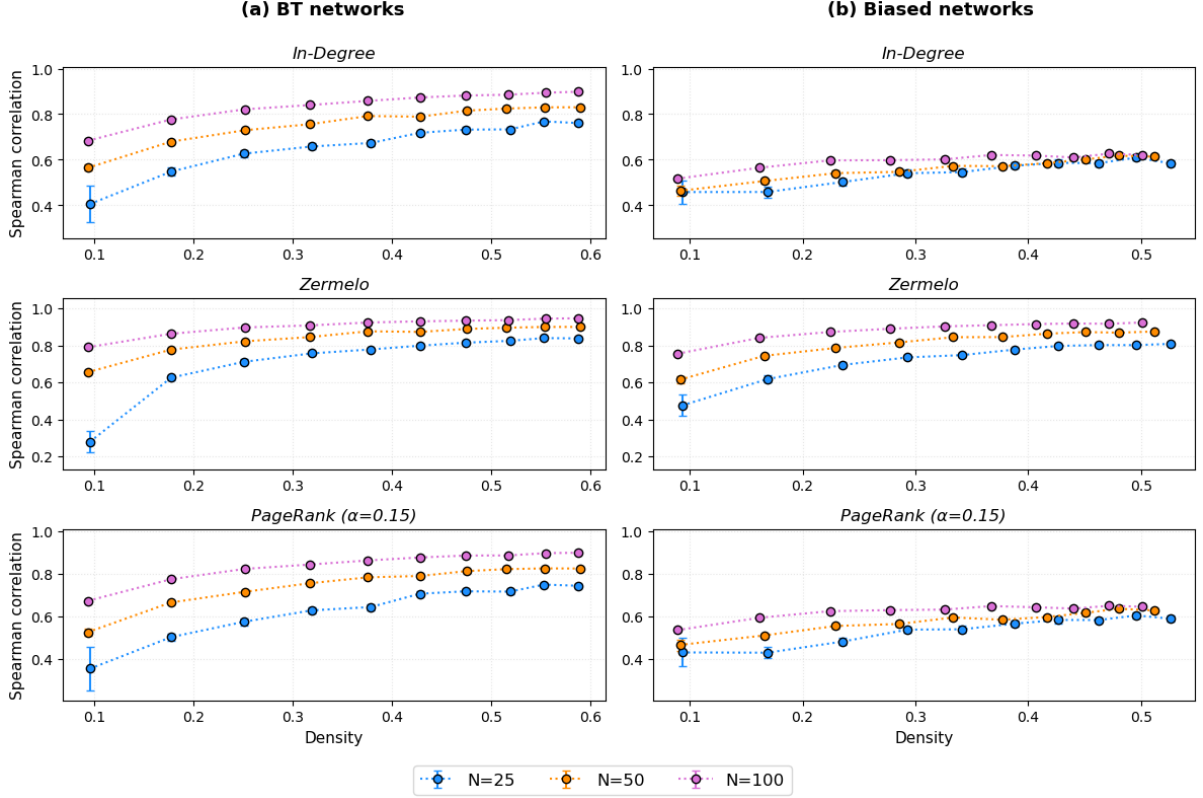


Figure 6: Spearman correlations between true intrinsic strengths and metric scores as a function of density d for different network sizes. The three rows of subplots present the results for in-degree, Zermelo and PageRank respectively. The curves within each plot show the comparison between $N = 25, 50, 100$.

5 Discussion

This section is focused on the analysis and interpretation of the results obtained from the numerical implementation. The discussion is structured to first evaluate the validity of the implementation of the metrics, then to compare the mutual relationships between the metrics, and finally to determine the influence of the network properties on their accuracy in relation to the intrinsic strength of the nodes.

5.1 Validation of Algorithmic Implementations

The first step in this analysis is to validate the accuracy and evaluate the efficiency of the implemented algorithms to ensure that they are functioning properly.

5.1.1 PageRank Validation and Efficiency

The iterative custom implementation of PageRank was validated by direct comparison with the function integrated into the `NetworkX` library. The average absolute difference between the scores obtained by the two methods being zero confirms the accuracy of the homemade function. However, an evaluation of the algorithms' speed reveals that

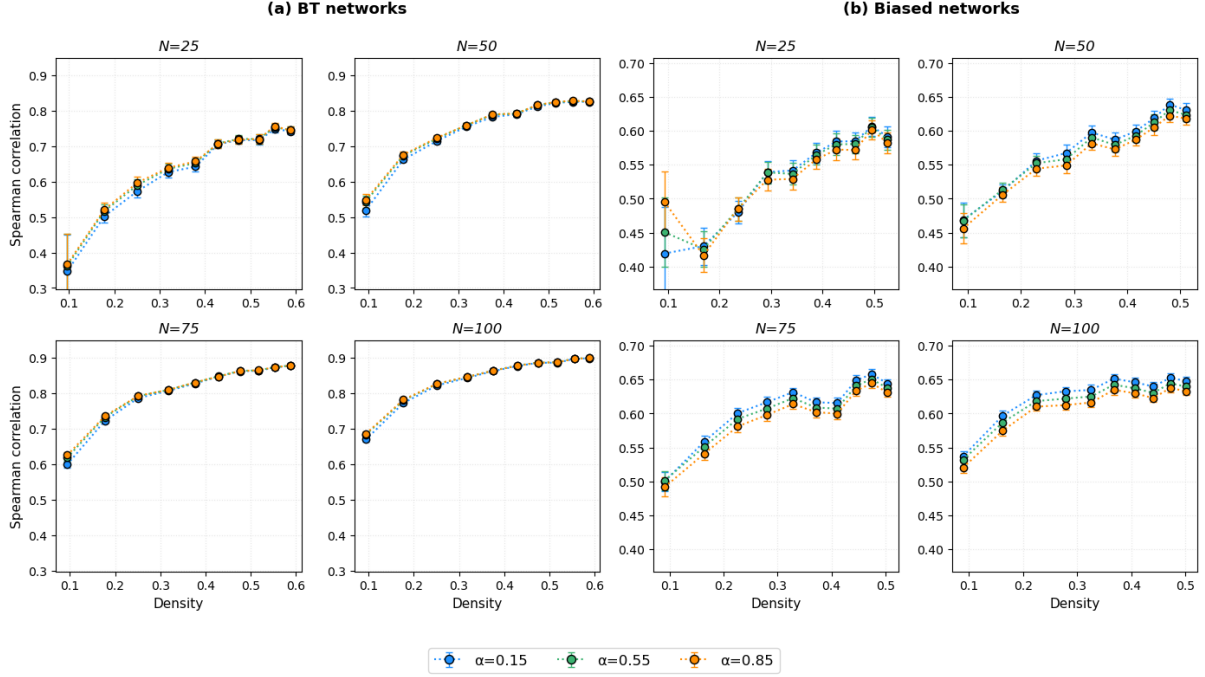


Figure 7: Spearman correlations between true intrinsic strengths and PageRank scores as a function of density d for different network sizes. The subplots show the correlation for four different network sizes: $N = 25, 50, 75, 100$. The curves within each plot represent different values of the teleportation probability ($\alpha = 0.15, 0.55, 0.85$)

the built-in function is more than five times faster. This difference can be explained by the fact that `nx.pagerank()` uses an optimised approach based on linear algebra, which allows the graph to be represented efficiently and avoids unnecessarily traversing all pairs of nodes [10]. Conversely, the homemade version uses loops, which are costly in terms of execution time. Consequently, the function `nx.pagerank()` was used for the rest of the project for reasons of convergence efficiency.

5.1.2 Convergence and Speed of Zermelo’s Strength

Zermelo’s strengths were calculated using an iterative procedure to achieve convergence. The results of Zermelo’s original algorithm (slow) and the faster version proposed by Newman (fast) were compared. Figure 1(a) clearly shows that the scores obtained by the two methods are very close, with all points aligned on the line $x = y$. This observation is confirmed by an extremely small absolute mean difference, of the order of 10^{-7} . This agreement is logical because both versions implement the same formula, the only difference being in the reorganisation of the terms of Zermelo’s algorithm to obtain the fast algorithm.

In terms of convergence speed, Figure 1(b) and Table 1 demonstrate that the fast algorithm is significantly more efficient. Considering the average number of iterations for each algorithm, the results are 7.66 ± 0.48 and 28.30 ± 3.19 , respectively. This means that Zermelo’s original algorithm requires approximately 3.7 times more iterations than the fast version.

Finally, the influence of the initial distribution on convergence was examined. The estimated final strengths remain identical, as confirmed by the very small differences shown in Table 2 and illustrated graphically in Figure 2(a) and (b). This is explained by the fact that Zermelo’s forces are estimated by MLE: since the likelihood function has only one maximum, this guarantees an identical final solution, regardless of the initial value. However, Table 1 shows that the uniform distribution is slightly faster than the other two initial distributions considered, for both algorithms. Figure 2(c) and (d) illustrates this trend graphically. One hypothesis to explain this difference in convergence speed is that the uniform distribution initially spreads the strengths, which reduces the variability between the initial values to zero and therefore the corrections needed at each iteration.

5.2 Correlations between metrics in Random Networks

This section examines the coherence between different ranking metrics when applied to simple random networks.

5.2.1 Correlation between In-Degree and Zermelo’s Strength

The correlation observed between in-degree and Zermelo’s strength is positive and moderate (0.666 and 0.679), indicating a significant but not perfect relationship. These values confirm that the more wins a node has (i.e., a high in-degree), the more it tends to have a high Zermelo’s strength. This reasoning is intuitive: winning often means being a strong participant.

However, the correlation remains far from 1. This is because the in-degree completely ignores the quality of the opponents. On the contrary, Zermelo’s strength relies on a model that gives more weight to wins against strong players. Thus, two nodes can have the same in-degree while having very different Zermelo’s strengths, depending on the opponents encountered.

Another interesting observation is that Spearman’s coefficient slightly surpasses Pearson’s one. This indicates that the order of the nodes is better preserved than the linear relationship between the values. In other words, in-degree is a more robust indicator for predicting a player’s rank than for estimating their actual strength.

5.2.2 Effect of the teleportation probability on PageRank Correlation with Other Metrics

The correlation between PageRank and in-degree is very high, ranging between 0.963 (0.970) and 0.978 (0.984) for Spearman (Pearson) depending on the teleportation probability α , as shown in Figure 3(a). This strong correlation can be explained by the fact that the PageRank algorithm is based directly on the redistribution of scores according to the number of incoming links: each entering link contributes to transferring a part of the score. As α increases, a larger proportion of the total score is redistributed evenly

to all nodes, which reduces the importance given to links and brings PageRank closer to a ranking based solely on the number of incoming links. This therefore increases the similarity with in-degree.

The correlation between PageRank and Zermelo’s strength remains weaker, at around 0.66 for Pearson and 0.67 for Spearman. It varies very little with α (see Figure 3(b)). This moderate correlation can be explained by the fact that PageRank diffuses the importance of a node according to its links and those of its neighbours, whereas Zermelo’s strength is a probabilistic estimate that takes into account the strength of the opponents defeated. Thus, in-degree has a better correlation with PageRank, as both metrics are based exclusively on link structure, whereas Zermelo takes into account the strength of defeated opponents and deviates from the logic of PageRank.

5.3 Metrics Performance Relative to Intrinsic Strength

This section looks at the accuracy of the three metrics by comparing them directly to the intrinsic strength of the nodes, as defined when generating Bradley-Terry networks and biased networks. The evaluation is still based on Pearson and Spearman correlation coefficients.

5.3.1 Comparison of In-Degree and Zermelo’s Strength

Zermelo strength is the most effective metric for estimating true intrinsic strength in both types of networks considered. Correlations are higher in all cases than those obtained with In-Degree, as can be seen in Table 3.

In-degree is particularly sensitive to the type of network. In BT networks, where each node participates in a comparable number of matches, in-degree shows a high correlation with true strength (between 0.6 and 0.7). This result can be explained by the fact that the total number of wins is a good indicator when all players face a certain number of opponents. However, as soon as participation becomes uneven, the performance of in-degree falls sharply (a drop of 26.7% and 34.0% for Pearson and Spearman respectively). This is because a mediocre player who has participated a lot can accumulate more wins than a strong player who has been used less. In this case, the in-degree becomes an unreliable indicator. Conversely, Zermelo’s strength remains high in both types of networks, with a decline of only 8% (Pearson) and 2.2% (Spearman). This shows that Zermelo remains effective, even in uneven configurations.

5.3.2 Impact of Teleportation Probability on PageRank

The influence of the teleportation probability α on PR performance is studied from Figure 4, for Bradley-Terry and biased networks. In BT networks, the correlation increases slightly as α increases, from approximately 0.766 to a maximum of 0.778 for the Pearson coefficient, at $\alpha = 0.85$. For higher values, the correlation tends to stabilise. Spearman’s

correlation follows the same trend, but shifted downwards. This evolution can be explained by the fact that PR is more sensitive to the structure of the graph when α is lower, which makes it sensitive to less active nodes. As α increases, a larger portion of the score comes from a uniform distribution across all nodes, which smoothes out these fluctuations and improves the correlation. However, if α becomes too large, the contribution of the graph structure gradually disappears in favour of the teleportation term, which reduces the possibility to differentiate participants' performances.

In biased networks, the trend is the same, but the correlations obtained are weaker for all values of α , with values between 0.490 and 0.521 (Pearson). This is due to the unequal participation of nodes, with some participants appearing much more often than others, regardless of their strength. As a result, heavily involved players may receive a high score even if they are not particularly strong, while high-performing but less involved players see their scores underestimated. In this case, the graph structure contains weakened information for comparing participants. As α increases, the teleportation term reduces the importance of this structure and gradually replaces this information with a uniform contribution that is not related to actual strengths. The PR score loses even more of the information it had, which explains why the correlation decreases as α increases in biased networks.

In summary, an increase in α corrects the small fluctuations caused by less active nodes in BT networks where the graph structure accurately reflects the hierarchy between participants. Conversely, it erases the source of information available in biased networks that already have a structure distorted by unequal participation, and therefore tends to decrease the correlation. It should be noted, however, that despite these differences, the dependence of correlations on α remains relatively weak in both models, with a variation over the entire interval of approximately 1.55% for BT networks and 6.13% for biased networks. This shows that the teleportation parameter plays a role in stabilising the score without completely changing the results. Thus, the choice of α does not have a big impact on the final scores. To sum up, the limitations of PageRank come from its strong dependence on the network structure rather than the teleportation parameter α .

5.4 Influence of Network Properties on Correlations

The previous section evaluated the correlations of the three metrics considered with intrinsic strength for a particular reference situation ($N = 50$ and $E = 100$). This section aims to evaluate the robustness of the metrics by analysing the dependence of the correlations on the network parameters. The influence of density, size, and teleportation probability α was studied in detail.

5.4.1 Influence of Network Density

In Bradley-Terry networks (see Figure 5(a)), correlation always increases with density for all three metrics. Zermelo maintains high values even for low densities, confirming its robustness and ability to estimate participant strength even when certain interactions

are missing. In-degree, on the other hand, shows high sensitivity to density: for low densities, it is unable to capture the entire hierarchy of participants, as it is limited to the number of direct wins. PageRank exhibits similar behaviour, with slight differences compared to in-degree. In biased networks (see Figure 5(b)), correlations are weaker for all metrics. PageRank and in-degree remain close to each other, but their performance is limited by unequal participation. PageRank still seems to perform slightly better in this case. Zermelo, on the other hand, maintains a higher and more stable correlation. It appears to be the best metric for handling participation biases and providing reliable estimates.

5.4.2 Influence of the Network Size

Figure 6 shows the effect of network size on correlations. In BT networks, an increase in the number of nodes slightly improves correlations for all metrics. For PageRank and Zermelo, a larger network reduces the variability in scores caused by poorly connected nodes. The in-degree also remains relatively stable. Indeed, when N is small, participants with few wins or links can have a large impact on relative scores. With a higher number of nodes, this effect is less influential. For biased networks, the effect of size is more limited. PageRank and in-degree are again sensitive to participation bias: in a small network, a few very active nodes can unbalance the scores, whereas this effect is mitigated in a larger network. However, network size alone is not enough to increase correlation. Zermelo, on the other hand, maintains stable performance and still appears unaffected.

5.4.3 Influence of the Teleportation Probability

Figure 7 examines PageRank as a function of the teleportation probability α for different sizes and densities. In BT networks, increasing α improves correlation minimally, reducing the impact of scores from inactive nodes while preserving relevant information. In biased networks, the opposite effect occurs: increasing α decreases correlation, as it homogenises scores and no longer takes graph information into account. This figure allows us to confirm the discussion made in subsection 5.3.2.

6 Conclusion

This project compared three ranking metrics in directed networks: in-degree, PageRank, and Zermelo’s strength. The analysis was conducted on different synthetic networks to evaluate each metric’s ability to estimate the intrinsic strength of nodes in various configurations. The results show that Zermelo’s strength is the most effective: it shows high correlations with intrinsic strengths in both types of networks studied. This robustness can be explained by the fact that Zermelo’s algorithm, based on the Bradley-Terry model, takes into account the quality of defeated opponents rather than the number of victories alone. In-degree, on the other hand, has significant limitations: it works well in Bradley-Terry-type networks where participation is uniform, but its performance drops sharply in biased networks. It does not take into account the strength of opponents and becomes unreliable when participants do not have a comparable number of participations. PageRank falls between the two. In BT networks, it correlates relatively well with intrinsic strengths, but remains much less effective than Zermelo. Its weakness is that it is highly dependent on the structure of the network and has difficulty distinguishing between truly strong participants and those who are simply very active. The properties of the network have a significant impact on the results. Moreover, network properties have a significant impact on results: higher densities and increased network size contribute to increased correlations.

This work was a first step towards the analysis of real-world professional tennis data as part of a Master’s thesis. One logical extension is to move from simple directed graphs to multigraphs, where the variable e_{ij} will be replaced by w_{ij} , the number of times player j defeated player i , thus allowing multiple confrontations. The main objective will be to develop a mathematical model that generates realistic synthetic tennis data reproducing patterns similar to real ATP results. It will combine programming to implement the model, data science to analyze real ATP data and analytical computations. This will help determine which ranking approaches are best suited to the specific structure of tennis competitions.

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